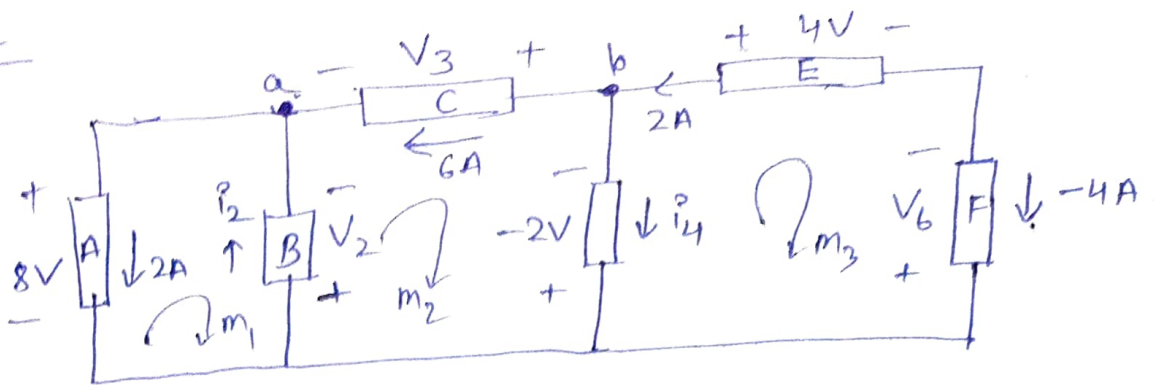




Using KVL in m_3 .

$$-(-2) = 4 - V_6$$

$$-V_2 = 8$$

$$V_2 = -8V$$

$$V_6 = 4 - 2 = 2V$$

①

Using KVL in m_2

$$-V_2 = -V_3 - (-2)$$

$$8 = -V_3 + 2$$

$$V_3 = -6V$$

①

At node 'a', apply KCL

$$i_2 + 6 = 2$$

$$i_2 = -4A$$

①

At node 'b', apply KCL

For 2A

$$2 = 6 + i_4$$

$$i_4 = -4A$$

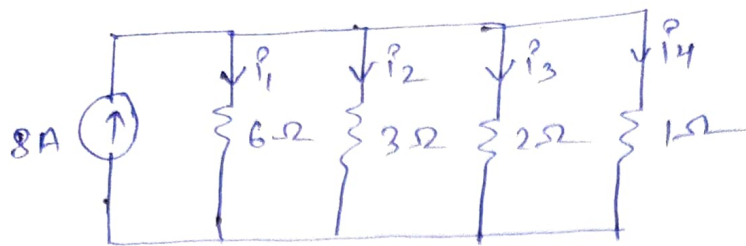
For -4A.

$$i_4 + (-4) + 6 = 0$$

$$i_4 = -2A$$

①

Q 2



$$i_1 = \frac{\frac{1}{6}}{\frac{1}{6} + \frac{1}{3} + \frac{1}{2} + \frac{1}{1}} \times 8A$$
$$= \frac{\frac{1}{6}}{\frac{2+4+6+12}{12}} \times 8$$

$$i_1 = \frac{2 \times 8}{24} = \frac{2}{3} A$$

$$\boxed{i_1 = \frac{2}{3} A}$$

①

$$i_4 = \frac{\frac{1}{1}}{\frac{1}{6} + \frac{1}{3} + \frac{1}{2} + \frac{1}{1}} \times 8A$$

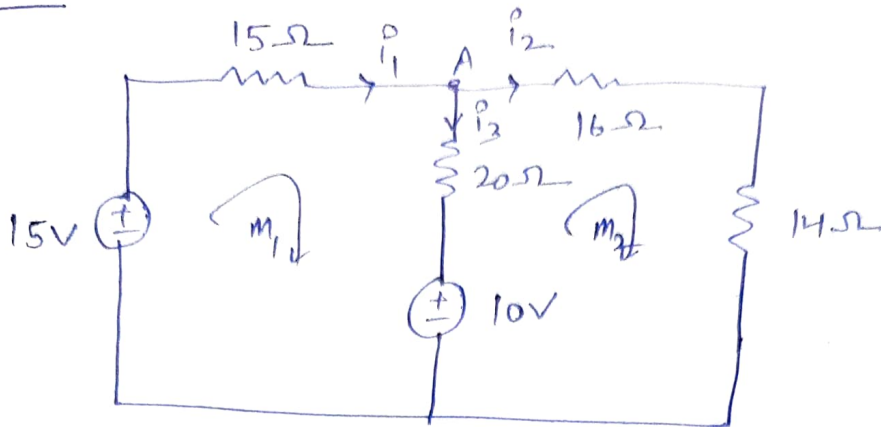
$$i_4 = \frac{\frac{1}{1}}{\frac{2+4+6+12}{12}} \times 8$$

$$i_4 = \frac{12 \times 8}{24} = 4$$

$$\boxed{i_4 = 4A}$$

①

Q3



Using KVL in m_1

$$15i_1 + 20i_3 + 10 - 15 = 0$$

$$15i_1 + 20i_3 = 5$$

$$3i_1 + 4i_3 = 1 \quad \text{--- (1)}$$

$$\left(\frac{1}{2}\right)$$

Using KVL in m_2

$$16i_2 + 14i_2 - 10 - 20i_3 = 0$$

$$30i_2 - 20i_3 = 10$$

$$3i_2 - 2i_3 = 1 \quad \text{--- (2)}$$

$$\left(\frac{1}{2}\right)$$

At node 'A', using the KCL.

$$i_1 - i_2 - i_3 = 0 \quad \text{--- (3)}$$

$$\left(\frac{1}{2}\right)$$

From (1), (2) & (3).

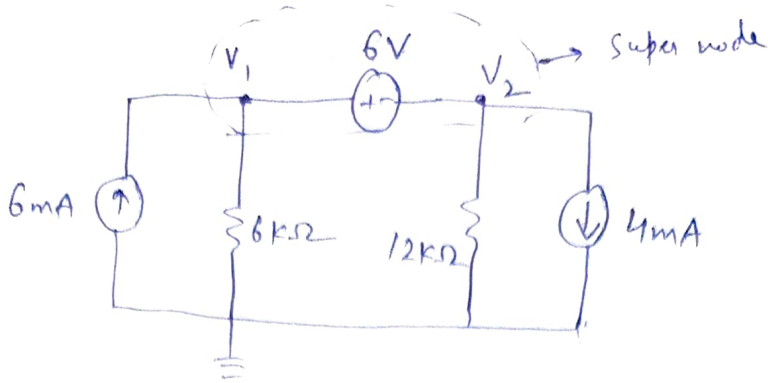
$$i_1 = \frac{1}{3} \quad ; \quad i_2 = \frac{1}{3} \quad ; \quad i_3 = 0$$

$$\left(\frac{1}{2}\right)$$

$$\left(\frac{1}{2}\right)$$

$$\left(\frac{1}{2}\right)$$

Q4



$$V_1 - V_2 = 6 \quad \text{--- (1)}$$

Applying KCL at Super node

$$6\text{mA} = \frac{V_1}{6\text{k}} + \frac{V_2}{12\text{k}} + 4\text{mA}$$

$$6\text{mA} = \frac{2V_1 + V_2 + 48}{12\text{k}}$$

$$72 = 2V_1 + V_2 + 48$$

$$2V_1 + V_2 = 24 \quad \text{--- (2)}$$

Add (1) & (2)

$$3V_1 = 30$$

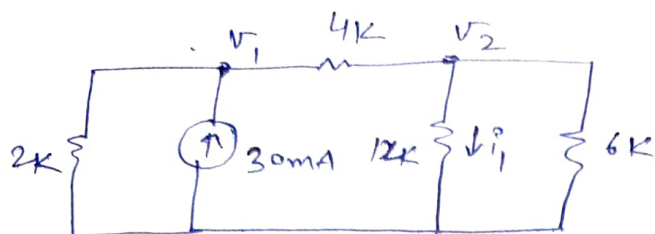
$$V_1 = 10\text{ V}$$

$$V_2 = 4\text{ V}$$

Q5

Case-I

Consider 30mA source



Using KCL at node A.

$$30m = \frac{V_1}{2k} + \frac{V_1 - V_2}{4k}$$

$$120 = 2V_1 + V_1 - V_2$$

$$120 = 3V_1 - V_2$$

$$120 = 3(2V_2) - V_2$$

$$V_2 = \frac{120}{5} = 24V$$

$$I_1 = \frac{V_2}{12k} = \frac{24}{12k} = 2mA$$

Using KCL at node B

$$\frac{V_2 - V_1}{4k} + \frac{V_2}{12k} + \frac{V_2}{6k} = 0$$

$$3V_2 - 3V_1 + V_2 + 2V_2 = 0$$

$$12k$$

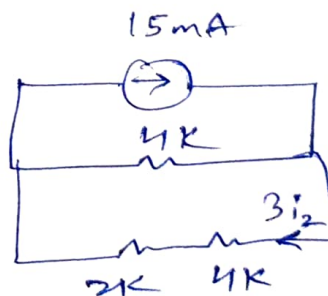
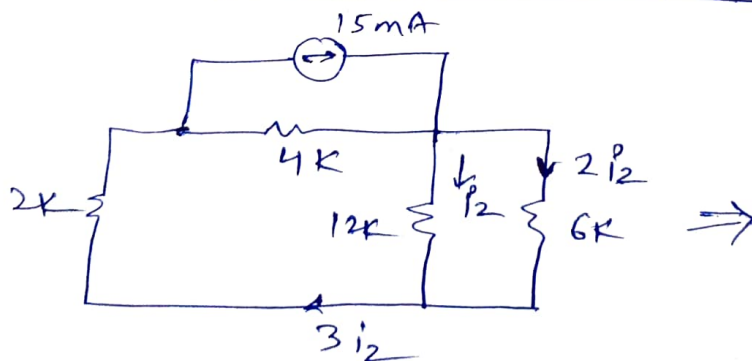
$$6V_2 = 3V_1$$

$$2V_2 = V_1$$



Case-II

Consider 15mA source



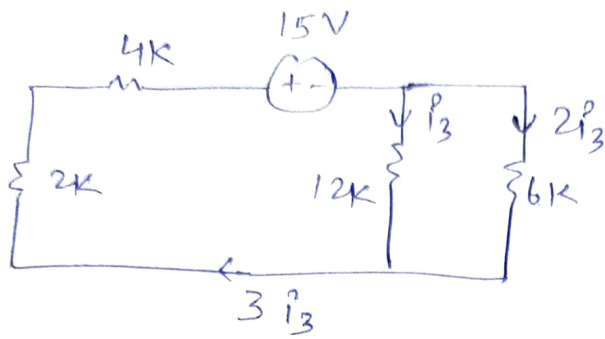
$$3i_2 = \frac{4}{4+6} \times 15m$$

$$I_2 = 2mA$$



Can-III

Consider 5V source



$$R_{eq} = (6K \parallel 12K) + 2K + 4K$$
$$= 10K$$

$$-3i_3 = \frac{15}{R_{eq}} = \frac{15}{10K}$$

$$i_3 = \frac{-15}{3 \times 10K} = -0.5mA$$

Using Superposition Theorem

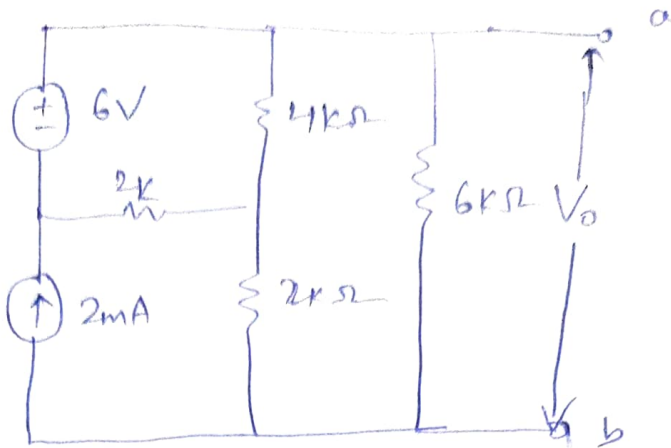
$$i^p = i_1 + i_2 + i_3$$

$$= 2mA + 2mA - 0.5mA$$

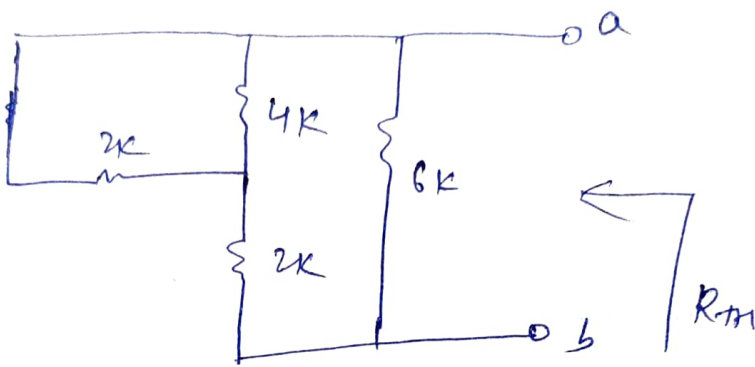
$$i^p = 3.5mA$$



Q6



Calculation of R_{Th}



$$R_{Th} = [(2 \parallel 4)k + 2k] \parallel 6k$$

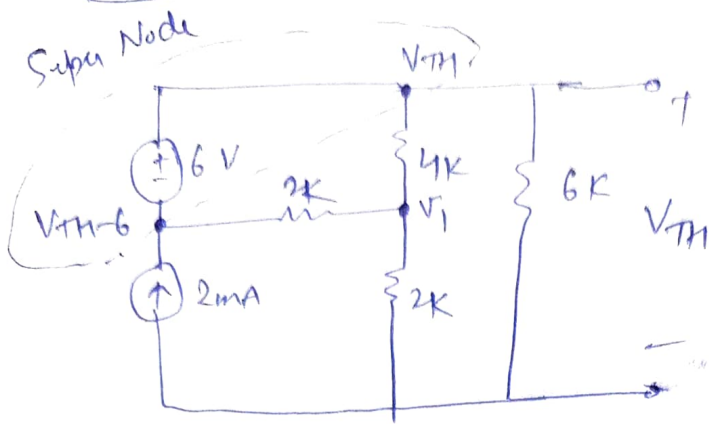
$$= \left[\frac{2 \times 4}{2+4} + 2 \right] \parallel 6k$$

$$= \left[\frac{10}{3} \parallel 6 \right] k$$

$$= \frac{\frac{10}{3} \times 6}{\frac{10}{3} + 6} k$$

$$= \frac{\frac{10}{3} \times 6}{\frac{28}{3}} k = \frac{10 \times 6}{28+47} = \frac{15}{7} k\Omega$$

Calculation of V_{TH}



KVL at ' V_1 '

$$\frac{V_1 - (V_{TH} - 6)}{2K} + \frac{V_1 - V_{TH}}{4K} + \frac{V_1}{2K} = 0$$

$$2V_1 - 2V_{TH} + 12 + \frac{V_1 - V_{TH}}{4K} + \frac{2V_1}{2K} = 0$$

$$5V_1 - 3V_{TH} = -12 \quad \text{--- (1)}$$

Applying KCL at super node

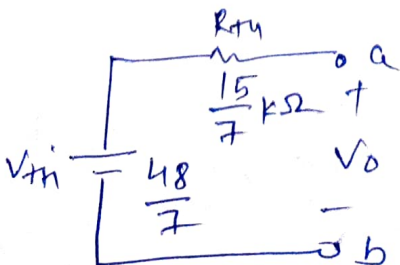
$$2m = \frac{V_{TH} - 6 - V_1}{2K} + \frac{V_{TH} - V_1}{4K} + \frac{V_{TH}}{6K}$$

$$24 = \frac{6V_{TH} - 36 - 8V_1}{2K} + \frac{3V_{TH} - 3V_1}{4K} + \frac{2V_{TH}}{6K}$$

$$11V_{TH} - 9V_1 = 60 \quad \text{--- (2)}$$

$$V_{TH} = \frac{48}{7}$$

The venin's Equivalent Circuit.



$$V_o = V_{TH} = \frac{48}{7} V$$

POSSESSION OF MOBILES IN EXAM IS UFM PRACTICE

Name.....

Enrollment No.....

Jaypee Institute of Information Technology, Noida
T1 Examination, ODD Semester 2024
B. Tech, 1st Semester

Course Title: Basic Electronics
Course Code: 24B11EC111

Max Time: 1 Hour
Max Marks: 20

CO1	Recall the concepts of various circuit elements and Kirchhoff's laws.
CO2	Understand the basics of semiconductor p-n junction diodes, op-amps and their applications.
CO3	Apply network theorems to solve complex DC networks.
CO4	Explain the operations of transistors (BJT and MOSFET) and analyze their biasing techniques.

Note: Attempt all the questions.

Q.1. Using Kirchhoff's laws, determine the values of i_4 , i_2 , v_3 and v_6 in figure 1.

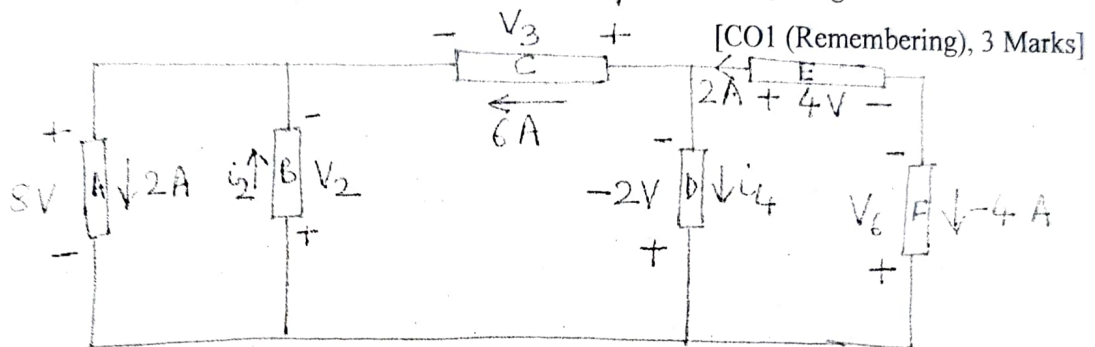


Figure 1.

Q.2. Use current division to determine the currents i_1 and i_4 in the circuit shown in figure 2.

[CO1 (Remembering), 2 Marks]

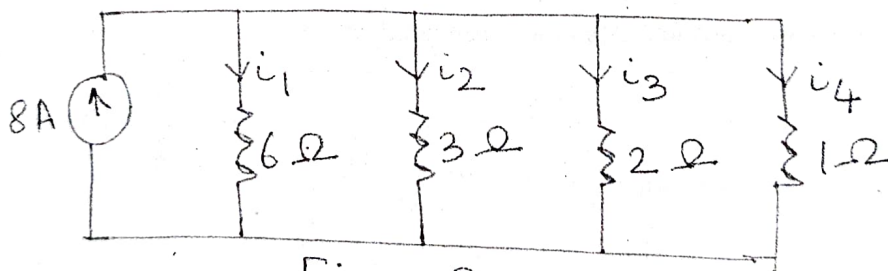


Figure 2.

Q.3. Find the branch currents i_1 , i_2 and i_3 using mesh analysis in figure 3.

[CO1 (Remembering), 3 Marks]

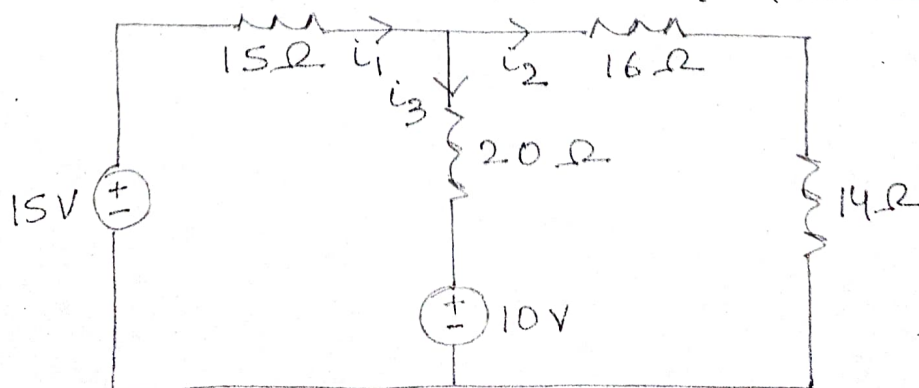


Figure 3.

Q.4. Determine node voltages (v_1 , v_2) for the circuit shown in figure 4.

[CO1 (Remembering), 3 Marks]

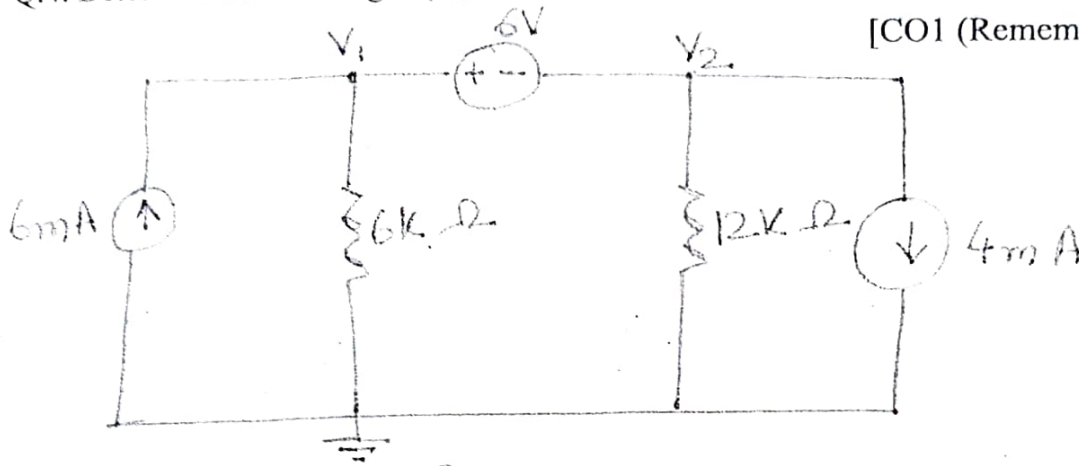


Figure 4.

Q.5. Use superposition theorem to find 'i' for the circuit shown in figure 5.

[CO3 (Applying), 4 Marks]

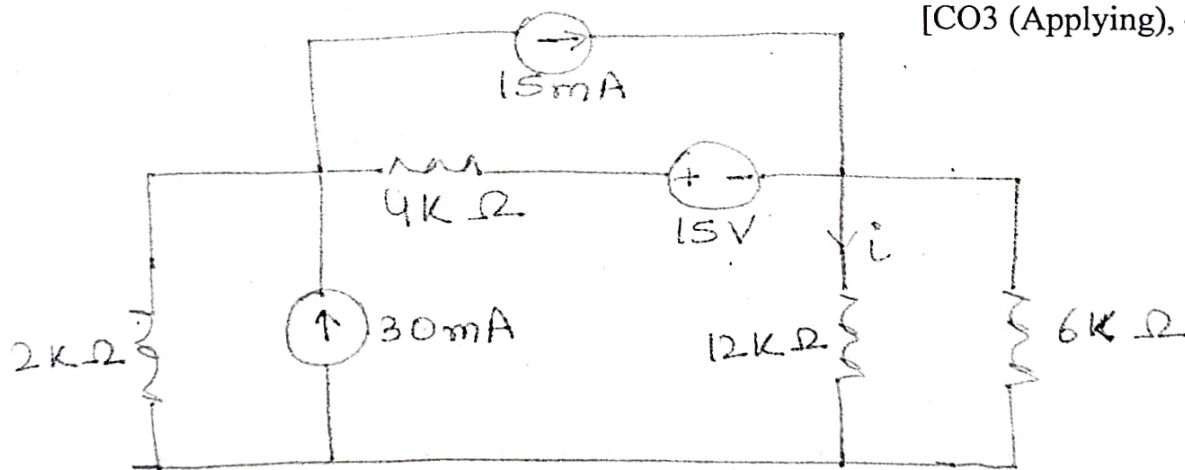


Figure 5

Q.6. Determine thevenin's equivalent circuit between terminal a and b of the circuit shown in figure 6 and find v_0 .

[CO3 (Applying), 5 Marks]

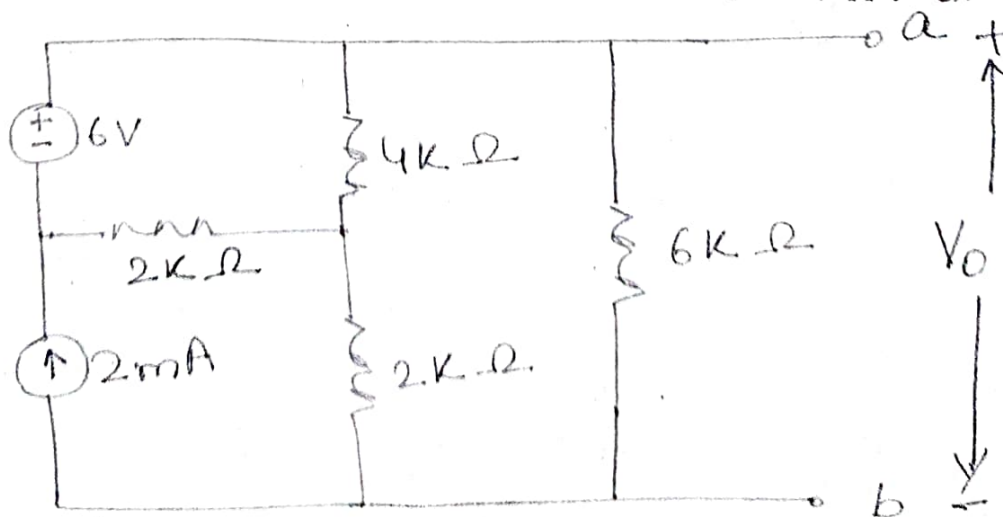


Figure 6.

POSSESSION OF MOBILES IN EXAM IS UFM PRACTICE

Name

Enrollment No.

Jaypee Institute of Information Technology, Noida
Test-2 Examination, ODD Semester 2024
B.Tech, I Semester

Course Title: Basic Electronics

Max Time: 1 Hour

Course Code: 24B11EC111

Max Marks: 20

Course Outcomes: At the end of the course, students will be able to:

CO1	Recall the concepts of various circuit elements and Kirchhoff's laws.
CO2	Understand the basics of semiconductor p-n junction diodes, op-amps and their applications.
CO3	Apply networks theorems to solve the complex DC networks.
CO4	Explain the operations of transistors (BJT and MOSFET) and analyze their biasing techniques.

Note: Attempt all the questions.

Q.1. (i). The current flowing through a p-n junction diode at room temperature is 100 nA when a large reverse bias voltage is applied. Calculate the current through the diode when 0.5 V is applied in the forward bias mode. Given $k = 1.38 \times 10^{-23} \text{ J/K}$, $T = 300\text{K}$.

[CO2 (Understanding), 2 Marks]

(ii). Derive an expression for efficiency of half wave rectifier.

[CO2 (Understanding), 2 Marks]

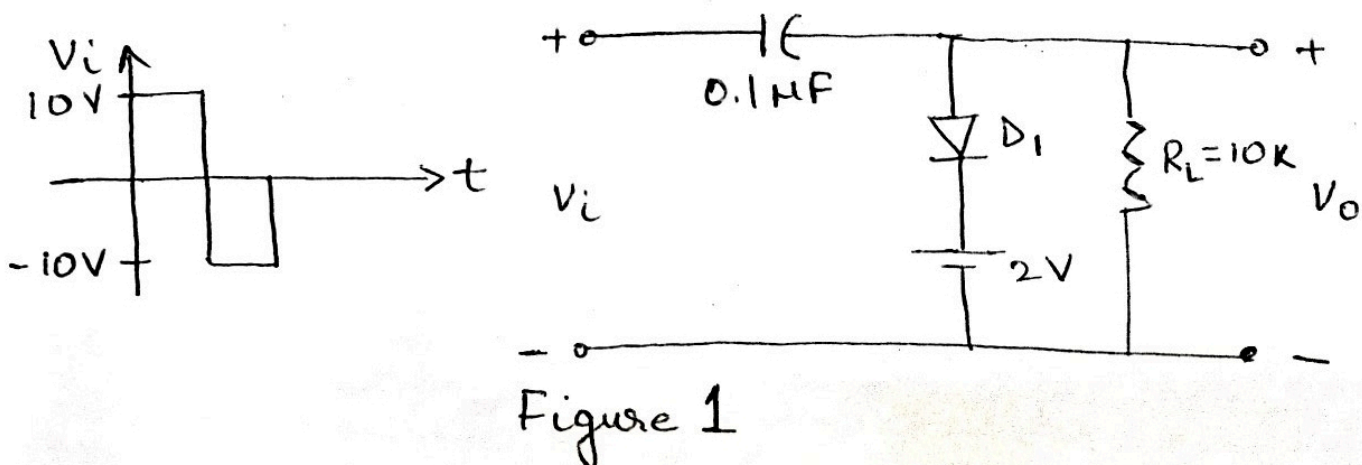
Q.2. A full wave rectifier is supplied with a sinusoidal voltage of 220 V at 50 Hz. The transformer connected to the rectifier has a turn ratio of $N_1 : N_2 = 10:1$. Assuming all the diodes to be ideal, determine the following:

- (i). DC output voltage.
- (ii). R.M.S output voltage.
- (iii). Form factor.
- (iv). Ripple factor.

[CO2 (Understanding), 1+1+1+1 Marks]

Q.3. Sketch the waveform for V_o for the equilibrium state of the circuit shown in figure 1. Consider the diode as Silicon diode ($V_T = 0.7 \text{ V}$). Solve it with the help of required steps.

[CO2 (Understanding), 4 Marks]



Q.4. Determine V_o for the circuit shown in figure 2. Consider the diode as Silicon diode ($V_T = 0.7$ V).

[CO2 (Understanding), 4 Marks]

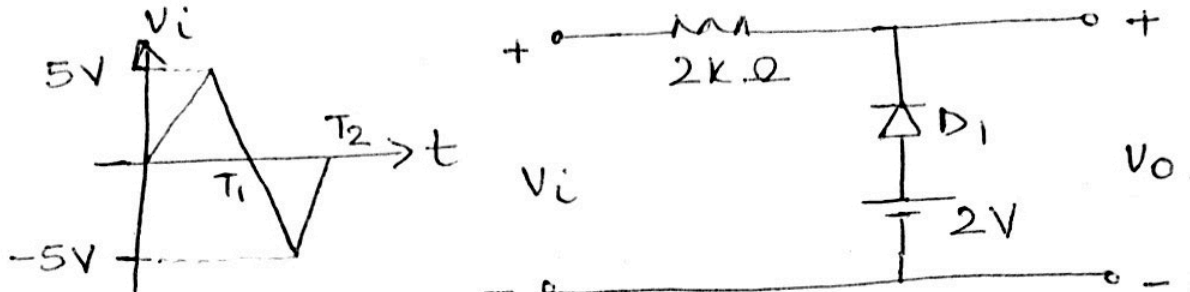


Figure 2.

Q.5. For the circuit shown in figure 3, determine the range of R_L and I_L that will maintain V_{RL} at 20 V. Also, calculate the maximum wattage rating required for the zener diode of the circuit.

[CO3 (Applying), 4 Marks]

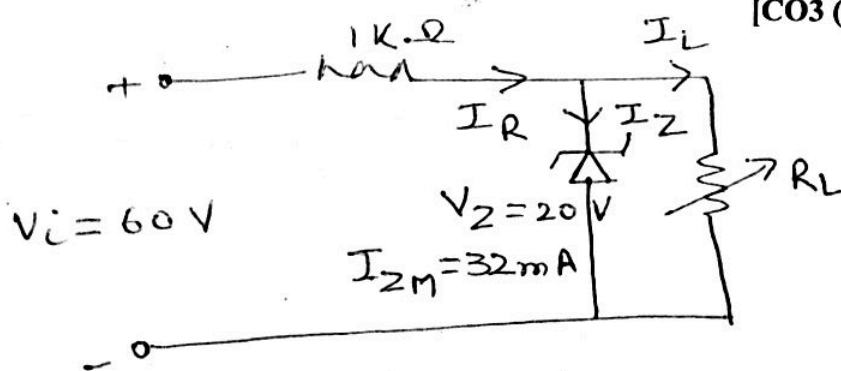


Figure 3

X X X — END OF PAPER — X X X

Solution 1 ^(a) T₂ Solutions. Assuming $\eta = 1$. (Si)

$$I_D = I_S \left[\exp \left(\frac{V_D}{V_T} \right) - 1 \right] \quad - (1)$$

Thermal Voltage, $V_T = \frac{kT}{q} = 26 \text{ mV} \quad - (\frac{1}{2})$

$$\therefore I_D = 100 \times 10^{-9} \left[\exp \left(\frac{0.5}{0.026} \right) - 1 \right]$$

$$\approx 22.48 \text{ A} \quad - (\frac{1}{2})$$

(b) Ratio of DC o/p power to the applied input a.c. power is known as rectifier efficiency. - (1)

Rectifier Efficiency $\eta = \frac{\text{DC o/p Power}}{\text{Input AC power}}$

$$= \frac{P_{dc}}{P_{ac}}$$

$$= \frac{I_{dc}^2 \times R_L}{I_{rms}^2 (R_L + r_f)} \quad - (\frac{1}{2})$$

$$= \frac{(I_m / \pi)^2 \times R_L}{(I_m / 2)^2 \times (R_L + r_f)}$$

$$= \frac{4}{\pi^2} \frac{R_L}{(R_L + r_f)}$$

$$= \frac{0.406}{\left(1 + \frac{r_f}{R_L} \right)} \quad - (\frac{1}{2})$$

Solution 2 → The question can be solved assuming full wave rectifier to be center-tap (OR) bridge. Hence, 2 solutions are provided.

①. Assuming Center-tap Rectifier.

$$V_{\text{secondary}} = \frac{1}{2} \left(\frac{V_{\text{primary}}}{N_1/N_2} \right)$$

$$= \frac{1}{2} \times \frac{220}{10} = 11$$

$$V_{m \text{ secondary}} = 11 \times \sqrt{2} \\ = 15.55 \text{ V.}$$

$$\text{ca) } V_{dc} = \frac{2 V_m}{\pi} = \frac{2}{3.14} \times 15.55 \\ = 9.90 \text{ V}$$

$$\text{cb) } V_{rms} = V_m / \sqrt{2}$$

$$= 15.55 / \sqrt{2} = 11 \text{ V}$$

$$\text{cc) Form factor} = \frac{V_{rms}}{V_{dc}} = \frac{11}{9.9}$$

$$= 1.11$$

$$\text{cd) Ripple factor} = \sqrt{\left(\frac{V_{rms}}{V_{dc}} \right)^2 - 1}$$

$$= \sqrt{(1.11)^2 - 1}$$

$$= 0.48.$$

②. Assuming it as a bridge rectifier.

$$V_{\text{secondary}} = \frac{V_{\text{primary}}}{N_1/N_2}$$
$$= \frac{220}{10} = 22 \text{ V}$$

$$V_{m \text{ secondary}} = 22 \times \sqrt{2}$$
$$= 31.11 \text{ V}$$

$$\text{ca) } V_{dc} = \frac{2V_m}{\pi} \quad - \left(\frac{1}{2}\right)$$
$$= \frac{2 \times 31.11}{3.14} = 19.82 \text{ V} \quad - \left(\frac{1}{2}\right)$$

$$\text{cb) } V_{rms} = \frac{V_m}{\sqrt{2}} \quad - \left(\frac{1}{2}\right)$$
$$= \frac{31.11}{\sqrt{2}} = 22 \text{ V} \quad - \left(\frac{1}{2}\right)$$

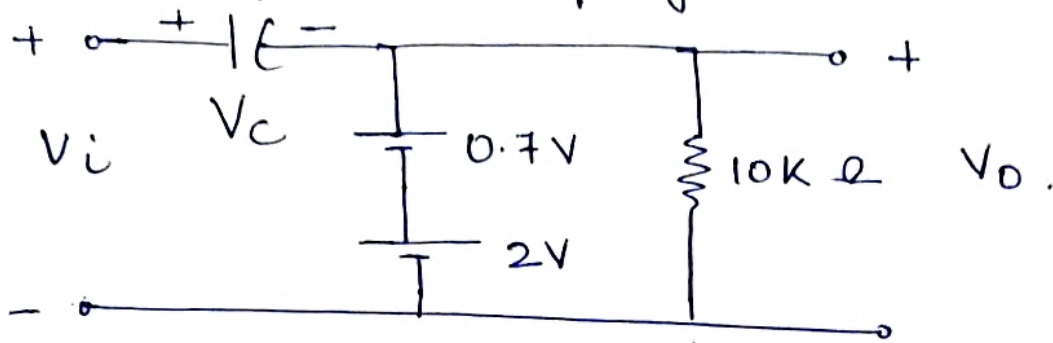
$$\text{cc) form factor} = \frac{V_{rms}}{V_{dc}} \quad - \left(\frac{1}{2}\right)$$
$$= \frac{22}{19.82} = 1.11 \quad - \left(\frac{1}{2}\right)$$

$$\text{cd) Ripple factor} = \sqrt{\left(\frac{V_{rms}}{V_{dc}}\right)^2 - 1} \quad - \left(\frac{1}{2}\right)$$

$$= \sqrt{(1.11)^2 - 1} = 0.48 \quad - \left(\frac{1}{2}\right)$$

Solution 3 →

Step 1 → for +ve half cycle



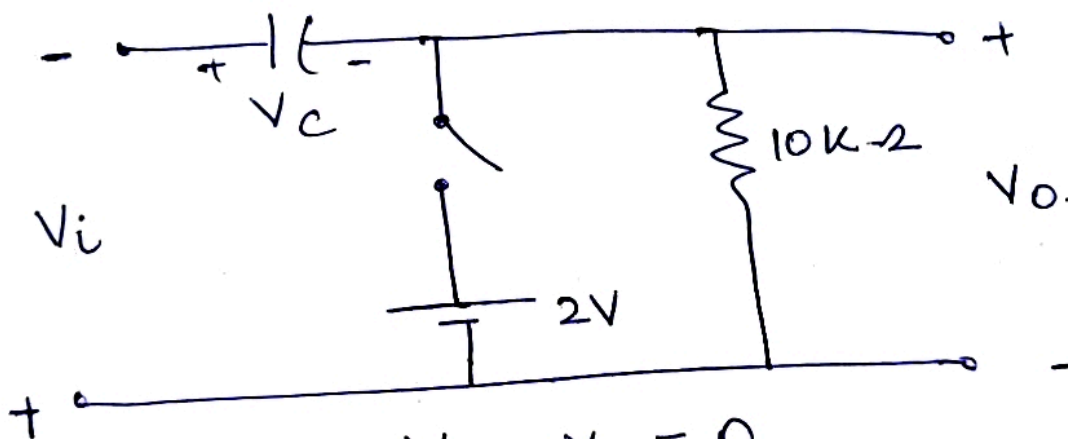
— (1)

$$V_o - 0.7 - 2 = 0$$

$$V_o = 2.7V$$

— (1/2)

Step 2 → for -ve half cycle.



— (1)

$$-10 - V_c - V_o = 0$$

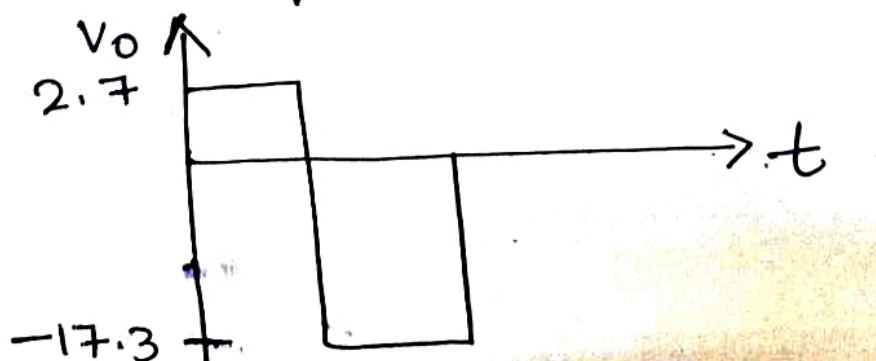
$$V_c = 10 - 2.7$$

$$\therefore -10 - 7.3 - V_o = 0$$

$$V_o = -17.3V$$

— (1/2)

$\therefore V_o$ waveform will be.



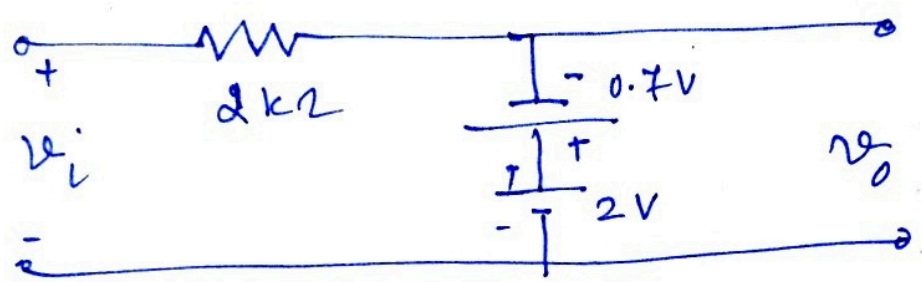
— (1)

Q4

For $0 \leq t < T_1$

Transitioning voltage can be determined by applying the condition, $I_f = 0$ at $V_D = 0.7V$.

For $V_i \leq 1.3V$, Diode is forward biased, hence, the resultant circuit will be as follows:

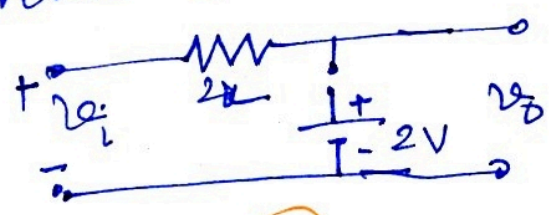


(1/2)

(1/2)

Hence, $V_o = 2 - 0.7 = 1.3V$.

For $V_i > 1.3V$, Diode is reverse biased, hence, the resultant circuit will be:



(1/2)

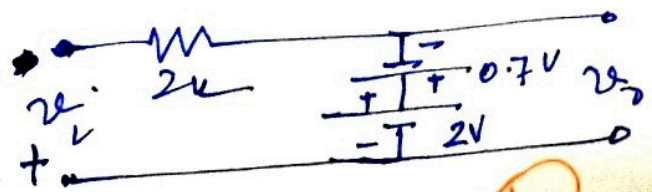
Thus $V_o = V_i$,

(1/2)

For $T_1 \leq t \leq T_2$: Diode is Forward biased for the entire duration, hence the resultant circuit will be:

Thus, $V_o = 2 - 0.7 = 1.3V$

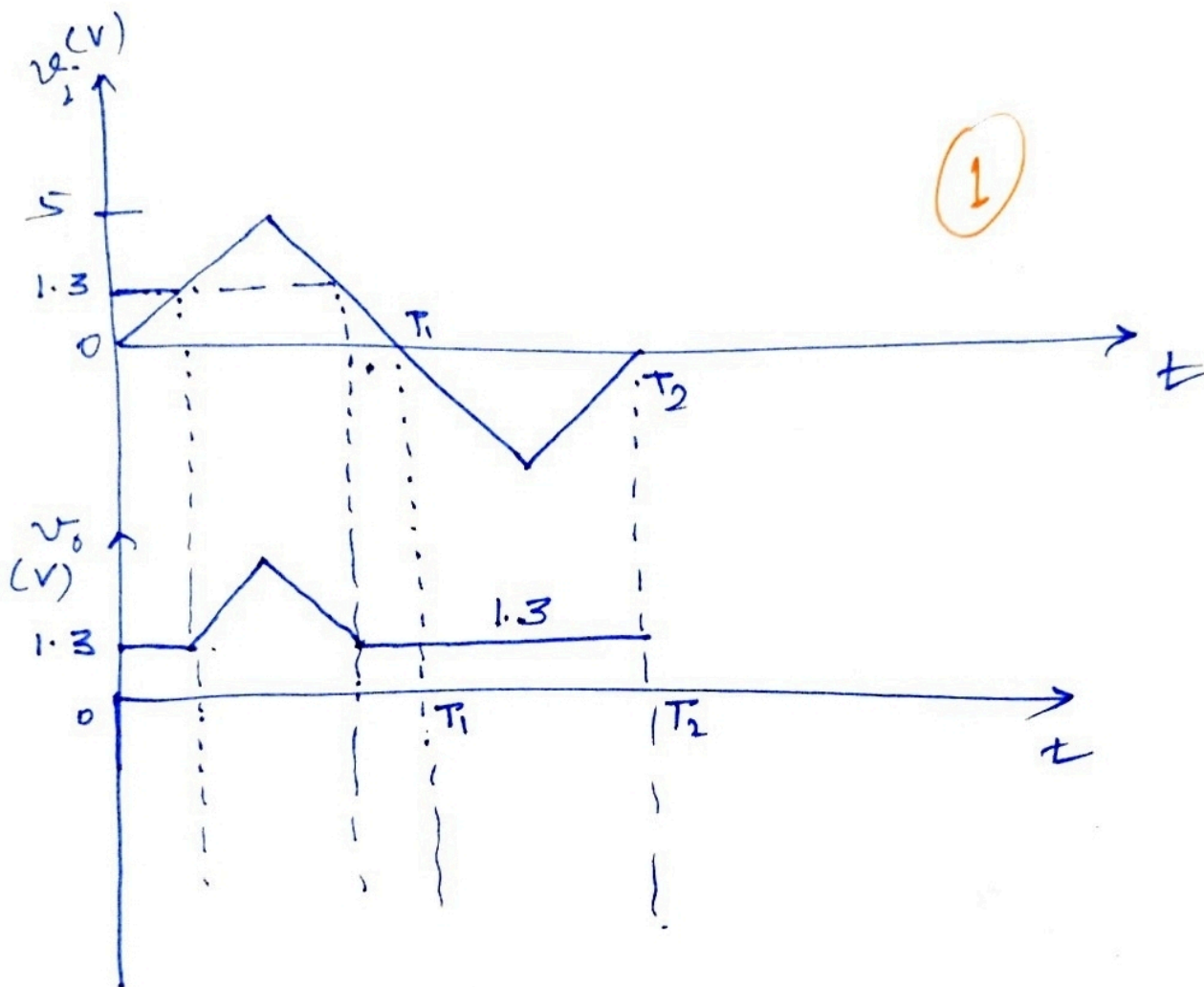
(1/2)



(1/2)

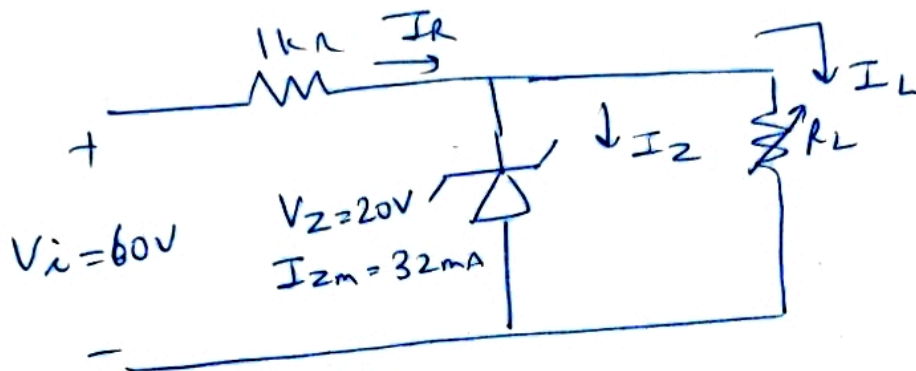
mod.)

Output waveform :



Q.5 For the circuit of Fig. 1, determine the range of R_L and I_L that will maintain V_{RL} at 20V. Also Calculate the maximum wattage rating required for the zener diode of the circuit.

(4)



Soln

$$R_{Lmin} = \frac{R V_Z}{V_i - V_Z} = \frac{1k \times 20}{60 - 20} = 500 \Omega$$

$$I_{Lmax} = \frac{20}{R_{Lmin}} = \frac{20}{500} = 40 \text{ mA}$$

$$V_R = V_i - V_Z = 60 - 20 = 40 \text{ V}$$

$$I_R = \frac{V_R}{R} = \frac{40}{1k} = 40 \text{ mA}$$

$$I_{Lmin} = I_R - I_{Zm} = 40 - 32 = 8 \text{ mA}$$

$$R_{Lmax} = \frac{V_Z}{I_{Lmin}} = \frac{20}{8 \text{ mA}} = 2500 \Omega$$

Range of $R_L = 500 \Omega$ to 2500Ω

Range of $I_L = 8 \text{ mA}$ to 40 mA

$$P_{max} = V_Z \cdot I_{Zm} = 20 \times 32 \text{ mA} = 640 \text{ mWatt}$$